

Multimodal Models for Skin Cancer Classification using Clinical Freetext and Dermatoscopic Images

Corresponding Author: Dr Matthew Watson

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

- Novelty here is the multimodal model skin cancer classification using dermatoscopic images, patient metadata, and clinical free text.
- Authors introduce a unique LLM-based pipeline to detect and remove "leading language" from clinical notes, mitigating bias that could lead to label leakage. I feel like this should be highlighted as a central novel component of the study
- unfiltered text yielding the highest AUROC (97.03%) is impressive
- The methodology is generally rigorous; statistical metrics are sound and repeated across multiple runs.
- Key limitations include the lack of public access to data/code and reliance on manual LLM prompting, which may affect reproducibility and generalizability.
- Clear value in potential real-world clinical impact, particularly in improving referral decisions in dermatology.

Reviewer #2

(Remarks to the Author)

The paper explores the extension of skin cancer classification models from vision-only approaches to multimodal systems that integrate dermatoscopic images, patient metadata, and clinical free text. The authors introduce a binary classification task (benign vs. malignant) based on secondary care outcomes using a newly collected multimodal dataset. A key contribution is a novel preprocessing pipeline that uses both regular expressions and large language models to filter out "leading language", i.e. text that may inadvertently leak diagnostic cues, at varying levels. Through systematic experiments, the study demonstrates that even filtered clinical text improves classification performance over image-only models, highlighting both the benefits and potential biases introduced by textual data. The findings underscore the value of incorporating clinical notes in multimodal learning, while emphasizing the need for bias-aware preprocessing in medical AI systems.

The paper addresses an important and timely topic by exploring the use of multimodal models, e.g. combining dermatoscopic images, patient metadata, and clinical free text, for skin cancer classification. The proposed text preprocessing pipeline, aimed at controlling bias from leading language in clinical notes, is a valuable contribution, and the reported performance improvements over image-only models are compelling.

However, several important concerns limit the current reproducibility and interpretability of the results. First, the absence of publicly available code significantly hinders the reproducibility of the approach, particularly given the complexity of the text filtering pipeline.

Second, while the authors report AUROC scores, the class imbalance in the dataset raises concerns about the actual performance of the models; confusion matrices are necessary to provide a clearer picture. In the current state, it is impossible in my humble opinion to assess the actual quality of the generated models. Additionally, explainability techniques should be applied to the obtained models to validate the plausibility of taken decisions.

Third, the examples of text filtering are presented in a somewhat opaque format using dashes, making it difficult to evaluate the effectiveness of the filtering process. Are the filtered words actually replaced by dashes (and it could potentially bias the

training) or is it presentation choice?

Lastly, the experiments are reportedly repeated only three times, and results are presented with mean and standard deviation. This is clearly insufficient to assess the robustness of the findings given the variability often observed in deep learning models.

In its current form, the paper offers interesting ideas but would benefit from clearer reporting, more comprehensive evaluation metrics, and increased transparency.

Reviewer #3

(Remarks to the Author)

This manuscript is the first of its type to evaluate the role of pre-processing of language from clinical records that is included as part of a multimodal evaluation of skin lesions with images, though medical NLP has been combining text and images for years. Clinicians have long recognized that there are many available, well-established risk factors for melanoma that can be superimposed on a visual examination of a specific lesion to improve diagnoses of malignancy. Creating an objective means of combining these factors in an unbiased way is a worthy and valuable contribution to the field. One philosophical point I think worth mentioning is that the authors seem to assume that a bias created by physicians in the language they use is a negative thing. However, in the complex world of medicine, there are still many intangibles and unknowns that are likely to give a provider (who has met the patient firsthand) greater insights (conscious and unconscious) into the likelihood of a malignant diagnosis beyond just the clinical examination. Further, it is not clear to me that even the Filt (full filter) will ever be able to completely remove language that represents bias. For example, even something as simple as the mention of sun exposure or of a particular skin type by a provider in a note is likely due to their feeling that this information is of particular help in making the diagnosis. This is different than removing non-factual statements because it reflects the provider's filtering of the information they choose to put in the medical record, which is not trivial. Even a full transcript of a provider-patient conversation would not be free of bias because the questions the provider asks is biased toward what they think are important given their interaction with the patient. Finally, in our era of augmented intelligence, it is likely that inclusion of some biases are likely very beneficial in improving diagnostic accuracy and the real challenge is going to be identifying how a provider perceives something concerning about the patient without knowing how they perceive it. This article is likely only a first step in the development of improved multimodal models for skin cancer identification that will tease out important and unimportant provider biases for improved diagnosis of skin cancer. Below are a few more specific comments:

1. The authors treat "clinically confirmed benign" diagnoses (visual inspection without biopsy) as equivalent to histopathologically confirmed cases, which is problematic dermatologist false negative rates of ~5% [their refs 48-51], and these community physicians are likely higher, yet they're using these potentially erroneous clinical diagnoses as ground truth for 93% of their "benign" cases. Is there any statistical modeling or analysis that could be done to assure that the results would be similar if there was a false negative rate of 8-10%, for example?

2. I did not see a comparison of their model to established benchmarks (without multimodal components or without pre-processing of language in multimodal models). Specifically, how did it compare to existing state-of-the-art models (e.g., those trained on ISIC 2020) or public data sets, or to dermatologist performance on the same dataset?

3. The surgical consultation notes inherently contain the clinician's diagnostic opinion, which likely informed the SNOMED-CT coding used as ground truth. Unless I am missing something, this creates a circular validation problem - essentially testing whether the model can extract the diagnosis that was already used to create the label. However, the comparison of all the components of the notes is extremely valuable.

4. Statistical questions: Are 3 runs with high standard deviations (e.g., $\pm 12.91\%$ for specificity in Table 6) adequate? I didn't see confidence intervals or significance testing between conditions or correction for multiple comparisons across numerous experiments. This could be easily addressed.

5. I would like to have a brief justification for using ConvNeXt-Large and BioClinicalBERT relative to other options. Is 5 epochs with a single learning rate adequate, what is your justification? Why was late fusion chosen over other possible multimodal architectures? Again, these are primarily justification/rationale issues that could be easily addressed.

6. I think it is critical to address some of the biases that exist in this work as a limitation and as a need for replication in other systems. Specifically, some of these that concern me are that the dataset comes from a single healthcare system's secondary care records (highly selected population that has already passed through primary care screening), making generalizability to primary care settings (their stated goal) questionable. It is also not clear how variable the images in the image data set are, how many dermoscopic images were available how many different providers and what their training might have been (probably not the same in different health systems, etc.).

7. Although most skin lesions evaluated (and even biopsied) are benign rather than malignant, with only 7% malignant cases, the model could achieve 93% accuracy by always predicting benign, which would represent little value to application of their model in the real world.

8. Is your model learning meaningful patterns or exploiting shortcuts going forward? Might be worth commenting on that.

9. Overlooked Key Literature: No citation of recent work on bias in medical AI (e.g., Obermeyer et al., Science 2019); Missing references to clinical text debiasing work beyond basic NER; No mention of recent multimodal medical imaging papers from

major conferences; Absent citation of work on spurious correlations in medical AI.

10. In the discussion I would like to see at least some discussion of potential failures of the system and when it should or should not be trusted, what the performance differences might be across different demographic groups, what regulatory requirements might be needed for clinical use, and how it might be integrated into clinical workflows. This could be a brief section, but will help guide future work.

11. The 97% AUROC with unfiltered text may be a success, but might be due to overfitting to textual shortcuts. The drop to 95.44% after filtering suggests the model might be primarily learning to extract diagnoses rather than learning from actual clinical features. This should be mentioned.

12. I am a little confused about what constitutes "leading language" and how that was validated – this seems to be a very subjective call. Inclusion of more specific rules and validations (perhaps in supplemental materials) might be really helpful.

13. Minor Issues: Table 1's comparison to other datasets is misleading as those datasets have verified pathological ground truth; the claim that regular expressions "cannot capture context" ignores decades of successful rule-based NLP; using SNOMED-CT codes as ground truth without discussing their accuracy and inter-coder reliability should be corrected.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

All my comments were adequately addressed by the authors

Reviewer #2

(Remarks to the Author)

The authors have provided most of the answers from my previous review. However, two main points need to be addressed in my honest opinion.

First, the confusion matrices provided in supplementary material should be more discussed. While not a lot of patients are missed in term of positivity, the (only) two analyzed models are making a lot of false positive errors. While the paper state that the model can be applied in practice, I am not sure how really applicable it is in the real life with that many mistakes. It could be interesting to compare these matrices with values that could be obtained by experts and non-experts of the pathology.

Additionally, the explainability provided in supplementary material should be carefully analyzed and not "dropped" as it is currently. When looking at the explanation provided in the examples, it appears, once again from my understanding, that the model are not focusing on the appropriate words (e.g. "gender" instead of the actual gender, "fitzpatrick" more that the type itself, etc.). Same observation with the image: why are the highlighted areas important for the decision making process? That is, first, not enough examples to actually asses the capacity of the model to take decision and second, not sufficiently analyzed to be sure the model is not biased in some way and actually taking the decision for the right reason. It is something central nowadays, in particular when the paper talks about applying the model in clinical settings.

Reviewer #3

(Remarks to the Author)

I am satisfied with the responses and changes made by the authors and find it acceptable for publication.

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors have answered to all my concerns.

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Multimodal Models for Skin Cancer Classification using Clinical Free Text and Dermatoscopic Images

Author’s Response to Reviewers

October 7, 2025

We thank all reviewers, and the editor, for their constructive feedback on our work. We are pleased to hear that all reviewers found our paper of great interest and value, and thank them for their positive comments that: our work introduces novel pipelines and has clear potential for real-world clinical impact (R1), has compelling evidence for the introduction of (filtered) freetext in medical AI systems (R2), and is the first of its type to evaluate clinical freetext in skin lesion classification (R3).

All reviewers have provided helpful comments from which we have amended parts of our work - this has only made our work stronger. Please see below for direct responses to comments from reviewers and details of how we have addressed these in our work. At the end of this document we also append our responses to the editor’s comments and hope our manuscript now fits the journal’s template.

Reviewer 1

Authors introduce a unique LLM-based pipeline to detect and remove "leading language" from clinical notes, mitigating bias that could lead to label leakage. I feel like this should be highlighted as a central novel component of the study

Author Response

We agree that this contribution should be made the main contribution of our study. We have revised parts of the introduction and the discussion to move the focus from our multimodal dataset and towards our leading language removal pipeline. We have also added more discussion of how this pipeline could be used outside of dermatology, hopefully inspiring wider use of our proposed techniques.

Key limitations include the lack of public to data/code and reliance on manual LLM prompting, which may affect reproducibility and generalizability.

Author Response

Both the data and the code are available upon request to the authors - the nature of the data requires us to closely monitor access to it. However, we have now added the LLM prompts used in the study to the supplementary material to aid reproducibility of this aspect of our work. We also hope that, by using open-source LLMs rather than closed-source models and APIs, this makes reproducing our work easier.

Reviewer 2

First, the absence of publicly available code significantly hinders the reproducibility of the approach, particularly given the complexity of the text filtering pipeline.

Author Response

As also noted above, code is available upon request to the authors. We've now also added the LLM prompts used for the text filtering to the supplementary materials so that people can easily reproduce this part of the pipeline.

Second, while the authors report AUROC scores, the class imbalance in the dataset raises concerns about the actual performance of the models; confusion matrices are necessary to provide a clearer picture.

Author Response

The reviewer is correct that AUROC is a poor metric for imbalanced tasks. Our results tables also report sensitivity and specificity, which we have now also added in the form of confusion matrices (found in the supplementary material) for easier visualisation. We have also added average precision as a metric, as this also is better suited for imbalanced data (as we now explain in the methods section).

Additionally, explainability techniques should be applied to the obtained models to validate the plausibility of taken decisions.

Author Response

We appreciate this suggestion from R2 and agree it would be beneficial to the paper. Unfortunately, in line with the data protection policy of the dataset we used, we cannot publish patient images. As a compromise, we have added some explainability plots to the supplementary material using samples from the ISIC 2020 dataset, although this comes with the caveat that it does not contain any freetext for our multimodal models. Nevertheless, we hope these plots aid the validation of our models.

Third, the examples of text filtering are presented in a somewhat opaque format using dashes, making it difficult to evaluate the effectiveness of the filtering process. Are the filtered words actually replaced by dashes (and it could potentially bias the training) or is it presentation choice?

Author Response

This is just a design choice we made for (what we believe) is easier visualisation of what is happening - the reviewer is correct that, if filtered words were replaced by dashes then training could still be biased. We have made this clearer where this is presented in the paper.

Lastly, the experiments are reportedly repeated only three times, and results are presented with mean and standard deviation. This is clearly insufficient to assess the robustness of the findings given the variability often observed in deep learning models.

Author Response

We have now ran all experiments 5 times (instead of only 3), and switched from reporting standard deviation to using 95% confidence intervals instead. We hope this gives a clearer picture of the robustness of our results. While in an ideal world we may wish to repeat experiments even more, we hope the reviewer agrees that, given we run more than 35 separate experiments for our work (each of which needs repeating separately), there is a point where this becomes impractical due to the cost and time required. We hope that this is an acceptable level of compromise, as it matches what is reported in similar deep learning studies.

Reviewer 3

One philosophical point I think worth mentioning is that the authors seem to assume that a bias created by physicians and the language they use is a negative thing. However, in the complex world of medicine, there are still many intangibles and unknowns that are likely to give a provider (who has met the patient first-hand) greater insights (conscious and unconscious) into the likelihood of a malignant beyond just the clinical examination.

Author Response

This is a very good point. We see the validity of including clinician-written text in an ML model for skin lesion classification (or, indeed, any medical classifier) as dependent on where in the clinical pipeline the AI tool is intended to sit. If the tool is intended as decision *support* to aid a clinician, then it may be suitable to include raw clinical text - as the reviewer points out, this likely includes clinical acumen not otherwise available. However, if the tool is intended as a decision *making* tool (i.e., somewhat replacing clinicians in actually making a diagnosis), then it is likely impractical to include this text in the model as it won't be available at inference time. In these situations, our proposed techniques can be used to make text a viable option.

We have added some text in the methods to explain this reasoning and motivation. In the discussion, we have expanded on how these tools could be used in different parts of the clinical workflow. We hope this makes these points clearer.

Further, it is not clear to me that even the FFilter (full filter) will ever be able to completely remove language that represents bias. For example, even something as simple as the mention of sun exposure or of a particular skin type by a provider in a note is likely due to their feeling that this information is of particular help in making the diagnosis.

Author Response

The FFilter filtering level is only designed to remove *clinical* leading language - text that would only really be able to be written after a consultation with a clinician. Language that could be patient responses to a simple questionnaire, such as sun exposure or skin type, are allowed as we believe these could be easily captured by the patient themselves. We have added a paragraph into the methods section to make this point clearer.

The authors treat "clinically confirmed benign" diagnoses (visual inspection without biopsy) as equivalent to histopathologically confirmed cases, which is problematic dermatologist false negative rates of 5% [their refs 48-51], and these community physicians are likely higher, yet they're using these potentially erroneous clinical diagnoses as ground truth for 93% of their "benign" cases. Is there any statistical modeling or analysis that could be done to assure that the results would be similar if there was a false negative rate of 8-10%, for example?

Author Response

The reviewer correctly points out that a small percentage of our benign ground truth labels may be incorrect. Unfortunately, however, we will always be unable to gain gold-standard histopathology results for these patients (in these cases, the risk of biopsy greatly outweighs any benefit). We are unsure if any extra analysis could be done to assure results would be similar given higher false negative rates in the ground truth - we do not know the real rate of false negative ground truth labels in our dataset, and so adjusting for this would be extremely difficult.

However, we don't believe this is a significant problem. The reason we only consider histopathologically confirmed malignant cases is as clinician false positive rates are significantly higher than their false negative rates, and this can be expected to have a significant effect on model performance. False negative rates, on the other hand, are much lower and so we expect benign labels to contain much less noise. While there will inevitably be some errors in the ground truth labels, this is always the case with labelled datasets; deep learning models have been shown to be robust to small amounts of label noise [1, 2], and so we do not expect this is affecting our models that much. Indeed, even histopathologically confirmed cases will be occasionally erroneously labelled.

Specifically, how did it compare to existing state-of-the-art models (e.g., those trained on ISIC 2020) or public data sets...

Author Response

This is something we wish we could have added to the paper, however there is unfortunately no fair comparison to be made using existing datasets as they do not contain any freetext. If we were to test our fully multimodal models on ISIC 2020, there would be no freetext for it to take advantage of. Conversely, while we could test existing models trained on ISIC 2020 on our own dataset, it would be impossible to know whether any difference in performance comes down to the freetext used by our model or a lack of generalisation to our data by the ISIC 2020 model. As such, we took the decision to omit any such comparison.

or to dermatologist performance on the same dataset?

Author Response

This would be a valuable addition to the evidence base for our techniques, but would be a whole study in and of itself to gain data from enough dermatologists for a statistically significant investigation. We have noted this as a direction for future work in the discussion.

The surgical consultation notes inherently contain the clinician’s diagnostic opinion, which likely informed the SNOMED-CT coding used as ground truth. Unless I am missing something, this creates a circular validation problem - essentially testing whether the model can extract the that was already used to create the label.

Author Response

In some cases, the surgery consultation notes may contain the clinician’s diagnostic opinion (which may or may not directly align with the SNOMED-CT code we extracted from elsewhere in the EHR for the ground truth). This is why our CFilt, DFilt, and FFilt freetext filtering techniques are so important, as these methods will remove this diagnostic data from the text. We note that our freetext filtering pipeline receives *only* the freetext to do this - no other parts of the EHR - so there is no circular validation problem here. We have added a small bit to the methods section making this clearer.

Statistical questions: Are 3 runs with high standard deviations (e.g., $\pm 12.91\%$ for specificity in Table 6) adequate? I didn’t see confidence intervals or significance testing between conditions or correction for multiple comparisons across numerous experiments. This could be easily addressed.

Author Response

This aligns with comments from other reviewers - as noted above, we have now run experiments 5 times. We have also switched from reporting standard deviation to reporting confidence intervals for our metrics.

I would like to have a brief justification for using ConvNeXt-Large and BioClinicalBERT relative to other options.

Author Response

This was already explained in Section 2.3, although we have slightly expanded upon this for clarity.

Is 5 epochs with a single learning rate adequate, what is your justification?

Author Response

This is standard practice for the models we use, and are the recommended hyperparameters by the authors of ConvNext and BioClinicalBERT. We have added a sentence and references pointing this out in the paper.

I think it is critical to address some of the biases that exist in this work as a limitation and as a need for replication in other systems.

Author Response

We agree that bias and generalisability are key concerns, not just in our work but all AI applications. In an ideal world, we would explore the performance of our techniques on external datasets, but unfortunately none exist with clinical freetext (that is the uniqueness of our work!). We do note that, while our data does come from only one healthcare provider, they do offer services across a wide geographic area which is reflected in the diversity of our dataset.

Nevertheless, there is more work to do here. We have noted this in the discussion, particularly around generalisability to other datasets.

Although most skin lesions evaluated (and even biopsied) are benign rather than malignant, with only 7% malignant cases, the model could achieve 93% accuracy by always predicting benign, which would represent little value to application of their model in the real world.

Author Response

While this is true for accuracy, this is not a metric we use in our paper (for this very reason). It could be argued that AUROC has similar problems, although as this is not a discriminative metric (i.e., it measures performance across all decision thresholds) it is not as pronounced as it is for accuracy. As discriminative metrics, we report sensitivity and specificity which are not susceptible to this issue. Furthermore, we have now also added average precision as a metric, as this is also a good indicator of model performance on imbalanced datasets. We have also switched to reporting AUROC in the range $[0, 1]$ rather than as a percentage, to try to avoid confusion with accuracy.

Is your model learning meaningful patterns or exploiting shortcuts going forward? Might be worth commenting on that.

Author Response

Our proposed freetext filtering pipeline aims to reduce the possibility of textual shortcuts as far as possible. As noted in the discussion of our manuscript, we argue that the reduction in classification model performance when FFilter freetext is used over Orig freetext demonstrates that this processing pipeline is successful. We also hope that the addition of the explainability plots to the supplementary material (as suggested by R2) further help to uncover the patterns learned by our models.

Overlooked Key Literature: No citation of recent work on bias in medical AI (e.g., Obermeyer et al., Science 2019); Missing references to clinical text debiasing work beyond basic NER; No mention of recent multimodal medical imaging papers from major conferences; Absent citation of work on spurious correlations in medical AI.

Author Response

Thanks very much for the recommendations - these have now been added in. We have also added more references to multimodal medical imaging, spurious correlations and expanded the literature review around clinical text debiasing.

In the discussion I would like to see at least some discussion of potential failures of the system and when it should or should not be trusted, what the performance differences might be across different demographic groups, what regulatory requirements might be needed for clinical use, and how it might be integrated into clinical workflows. This could be a brief section, but will help guide future work.

Author Response

We have added a bias analysis across the demographic data available in our dataset, and discussed this in the results section. We have also modified the discussion to make explicit suggestions around how different parts of our technique could be used in different parts of clinical practice, alongside the limitations of our work that need to be addressed before it can actually be deployed.

The 97% AUROC with unfiltered text may be a success, but might be due to overfitting to textual shortcuts. The drop to 95.44% after filtering suggests the model might be primarily learning to extract diagnoses rather than learning from actual clinical features. This should be mentioned.

Author Response

We completely agree with the reviewer’s conclusions; we apologise if this was not clear in the original manuscript. We have adjusted the results section to make this clearer. We also hope that the additions we made for the above point (adding in where this could be used in the clinical workflow) aids this, as it explores how even the unfiltered text models may be useful despite the bias shown.

I am a little confused about what constitutes “leading language” and how that was validated – this seems to be a very subjective call. Inclusion of more specific rules and validations (perhaps in supplemental materials) might be really helpful.

Author Response

We have now added the LLM prompts used for this to the supplementary material. We hope this, combined with our response to R3’s second point, addresses this issue.

using SNOMED-CT codes as ground truth without discussing their accuracy and inter-coder reliability should be corrected.

Author Response

We have added a discussion of the issues with SNOMED-CT to section 2.1.2. As now noted in the paper, we believe that most issues with the accuracy of SNOMED-CT codes are alleviated as we further group them into ‘benign’, ‘malignant’, and ‘ambiguous’ classes.

Minor Issues: Table 1’s comparison to other datasets is misleading as those datasets have verified pathological ground truth;

Author Response

We have double checked the datasets in Table 1, and find that most (if not all) of those datasets are similar to ours - they have histopathological ground truths for malignancies but not benign cases. This aligns with what we would expect from clinical practice - comparatively few patients with benign conditions will undergo biopsy due to the risks and costs involved (generally, the only histopathologically confirmed benign cases you would expect should be false positive clinical diagnoses).

the claim that regular expressions “cannot capture context” ignores decades of successful rule-based NLP

Author Response

Although we agree with the reviewer that the claim ‘cannot capture context’ ignores a lot of successful work, we have double checked the claim made in the paper and it is not as strong as this. We believe our weaker claim of ‘difficult to generalise to concepts’ stands.

References

- [1] David Rolnick, Andreas Veit, Serge Belongie, and Nir Shavit. Deep learning is robust to massive label noise. *arXiv preprint arXiv:1705.10694*, 2017.
- [2] Gökrem Algan and Ilkay Ulusoy. Label noise types and their effects on deep learning. *arXiv preprint arXiv:2003.10471*, 2020.

Multimodal Models for Skin Cancer Classification using Clinical Free Text and Dermatoscopic Images

Author's Response to Reviewers - Round 2

December 18, 2025

We thank all reviewers again for their time in reviewing our manuscript - it has truly made the work stronger. We are grateful for all reviewers re-reviewing our revision, and are happy to see R1 and R3 believe we have addressed all of their concerns. We have made a second round of revisions to the manuscript, addressing R2's remaining concerns and the points raised by the editors. Below, we first directly respond to R2 and indicate where we have improved the manuscript based on their constructive comments. We then identify where the manuscript has changed to address the editor's comments.

Reviewer 2

First, the confusion matrices provided in supplementary material should be more discussed. While not a lot of patients are missed in term of positivity, the (only) two analyzed models are making a lot of false positive errors. While the paper state that the model can be applied in practice, I am not sure how really applicable it is in the real life with that many mistakes. It could be interesting to compare these matrices with values that could be obtained by experts and non-experts of the pathology.

Author Response

This is an issue when the benign caseload is so much larger than the malignant caseload - even with high sensitivity and specificity values, one can end up with a high 'raw' number of false negatives/positives (in our case, positives, due to our decision to favour sensitivity). This is also the case in clinical practice, however, with both the sensitivity and specificity of our models being higher than those reported by primary care clinicians. As such, our model could still be used to improve the referral efficiency of primary care to dermatology. We have added a paragraph on this in the discussion.

Additionally, a NICE review of comparable digital technologies for the detection of melanoma highlights the performance of Skin Analytic's DERM. DERM is an MHRA and EMA class III regulated model which has false positive rates of 36% when sensitivity is fixed at 95% (<https://www.nice.org.uk/advice/mib311/chapter/Clinical-and-technical-evidence>). DERM is already being used in real-world clinical practice, despite its higher false positive rate than the model we describe in the paper, further highlighting the clinical utility of our proposed methods.

Furthermore, we would also like to note that, as our models achieve high AUROC and AP scores, this shows that it is possible for future work to easily tune the decision threshold of our models to achieve a high specificity, instead of a high sensitivity. Although this may come with slightly more false negatives, false positive numbers would be greatly reduced which may be preferable in settings where benign cases vastly outnumber malignant. For example, under such a setting, the model could be used to make auto-referrals for biopsy while all other patients remain examined by primary care. We have now also mentioned this in the discussion.

We completely agree that it would be extremely valuable to directly compare model performance with clinician performance. However, as we had already stated in the discussion/conclusion, this is unfortunately outside the scope of the study; this would be a large prospective study in and of itself, that would entail a lot of clinician time. As such, we leave this as future work, as has now been emphasised in the manuscript.

Additionally, the explainability provided in supplementary material should be carefully analyzed and not "dropped" as it is currently. When looking at the explanation provided in the examples, it appears, once again

from my understanding, that the model are not focusing on the appropriate words (e.g. “gender” instead of the actual gender, “fitzpatrick” more that the type itself, etc.)

Author Response

We have added an analysis of these results in the Results section. We hope this makes it clearer - in particular, we want to note that although some feature attribution is placed on the feature names, this can be explained due to the way transformers and the explainability methods work, and that there is still a lot of overall attribution placed on the other areas of freetext. Please also see the below response to R2’s final comment, which is also about explainability.

Same observation with the image: why are the highlighted areas important for the decision making process? That is, first, not enough examples to actually asses the capacity of the model to take decision and second, not sufficiently analyzed to be sure the model is not biased in some way and actually taking the decision for the right reason. It is something central nowadays, in particular when the paper talks about applying the model in clinical settings.

Author Response

Again, as above, we have added an analysis of the explainability samples we provide. This additional discussion highlights that, by and large, the image explainability also aligns with clinical reasoning.

However, as we noted in the previous response to reviewers and in the manuscript itself, we are limited to what publicly available data is out there. As no publicly available dermatoscopic datasets contain freetext (and we are unfortunately unable to release our own samples in the paper due to patient privacy regulations), we have had to create synthetic freetext (and sometimes metadata) to go alongside the ISIC images used. We believe this limits the usefulness of the explainability analysis, as it will add in certain artefacts not present with real data - and is the reason we have decided to include it as supplementary information, rather than the main text. There will also be some data shift between ISIC images and our own. As such, adding in more explainability samples may not be as helpful as one would hope and, as this response has shown each sample would have to be analysed in-text, we believe it would be difficult to accomplish without the whole paper becoming about explainability.

Furthermore, explainability methods themselves are unable to accurately identify biases in models (hence why we also include our own bias analysis of the results across age, sex, and Fitzpatrick skin types) as the reviewer suggests. They are also only designed to shed light on features used by models - and cannot be used to definitively say they always use certain features. These are limitations of all explainability techniques currently available in the literature. Overall, while we agree that explainability complements the existing model results and discussion, unfortunately it is not a ‘silver bullet’ and we have identified this in the discussion, along with how it might affect deployment in clinical practice.

Editor and Assistant Editor

Include the cohort details (sample size, etc) in the Methods part of the Abstract.

Author Response

These numbers are now added to the methods part of the abstract.

Keep consistent the use of ‘free text’ or ‘freetext’ throughout the manuscript.

Author Response

All instances of ‘free text’ have been replaced with ‘freetext’.

The term ‘Novelty’ is used twice in the manuscript, please modify.

Author Response

The two claims of novelty have been removed.

Remove the numbering from the headings/subheadings.

Author Response

Numbering has been removed, and we have double checked that any in-text references to sections now refer to the section name rather than the number.

You need to add a sentence to the Data Availability statement to describe how source data can be accessed using text similar to “The source data for Figure X is in Supplementary Data Y”. It needs to be specific about which figures you provide source data for, and specifically where it can be found.

Author Response

We have added that the provided CSV file contains data for Figure 5 and Tables 4-6. All other figures report overall patient statistics.

Add a figure legend for all Figures and include the sample size, where relevant.

Author Response

The caption for Figure 5 already includes sample size (5 runs of the model), and we have added it to Figure 4. Sample size is not relevant for other figures, as they report dataset statistics. We believe all figures include legends where needed - either in the Figure itself, in its caption, or both. Please let us know if there is a specific figure causing issues.

We only allow 10 display items (figures/tables), please move the remaining display items to the Supplementary Material and update the labels/ in-text citations.

Author Response

We have moved what were originally Figures 2 and 3 in the main text to supplementary material, and updated the in-text references accordingly. We are now at exactly 10 display items.

The Appendix A in the Supplementary Material on page 1 will need a header of either ‘Supplementary Methods’ or ‘Supplementary Results’. Text in the Supplementary materials must be cited in the main text. We encourage any key text to be moved to the main text.

Author Response

We have changed the name of this section, and ensured it has been cited in the main manuscript. We have decided to keep this text in the supplementary material - it is simply a definition of multimodal data, as we are aware it may have different definitions between different fields (e.g., computer science/medicine), however we expect the vast majority of readers to be aware of its meaning in this context.

The Llama Prompts in the Supplementary Materials pages 2-3 will need to be converted into Figures or Boxes and labelled/cited accordingly.

Author Response

We have changed these to boxes.

Panels a and b are not described in the legend of figure 3, 6.

Author Response

This has been fixed.

Please provide a statement confirming the cited (original) study has IRB approval, participant consent and permission for secondary use. If the need for informed consent was waived by the ethics committee for this study, please state this and the reason clearly.

Author Response

A couple of sentences explaining the IRB process followed has been added to the ethics statement.